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# MANISH KUMAR

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## Professional Summary

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### Development Experience

- Last worked at Encora Inc. as a Technical Lead Engineer on Media production platform - Avid Content Core (ACC) where my role is to develop Python service module and contribute to microservices design and architect.
- Seasoned Python developer with a career spanning over 10 years in Designing, Development, Architect, DevOps, and Cybersecurity in verticals like Healthcare, Telecom, Cybersecurity and Media Production.
- Experience in development of multitude of Python backend applications, REST APIs, Microservices, Database with cloud technology and network & security applications for Windows, Linux and Raspberry Pi.
- Experience in tools and technologies like Django, FastAPI, Kubernetes, Docker, Helm, CICD, DevOps, GitLab, Tilt Dashboard, Snyk security, Nix environment.
- Experience in cloud technologies like AWS EC2, EKS, SES, SQS, Lambda, RDS, S3, API Gateway, CloudFormation.
- Experience in developing applications based on Clean Architecture, MVT, layered and microservices architecture.
- Experience in creating test cases in Python using unit test and PyTest and doctest and worked with PyLint to analyze code for errors, standards and improvements.
- Experience in designing database tables, Django modeling, writing SQL queries and managing database backup-restore on Postgres DB.
- Experience in integrating GenAI tools like ChatGPT to backend with FastAPI and familiar with Agentic AI, Agents, RAG, AI Tokens.
- Proficient in utilizing LLMs and GenAI tools for advanced code generation, optimized debugging, and accelerating software development lifecycles.

### Security Experience

- 4 years of experience in designing security platforms, vulnerability scanner, security profiling, packet analyzer, threat detection and prevention for endpoints, servers and network monitoring.
- Experience in developing custom security solutions using Python with integration of security tools like Scapy, Nmap, Snort, NetFilterQueue, IPtables to assess network and system vulnerabilities, storing data and generating reports using Matplotlib and ReportLab.
- Experience in cybersecurity also includes analyzing security threats in systems, analyzing network vulnerabilities, Operating system hardening, threat prevention, system events, permissions, IP hashing.

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## Technical Skills

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**Programming:** Python, SQL, shell scripting

**Databases:** Postgres, MySQL, MongoDB

**Frameworks:** Django, FastAPI, Scapy, Django Rest Framework

**Python Module:** Twisted, Swagger, ReportLab, PyLint, Pyinstaller, Matplotlib

**Security Tools:** Nmap, Tcpdump, Wireshark, iptables, NetFilterQueue, Yara, Ghidra, Snyk, Snort

**Cloud platforms:** AWS EC2, Lambda, SES, RDS, ESC, SQS, EKS, CloudFormation

**Misc:** Docker, Kubernetes, CICD, AWS, Postman, OAuth, JWT, Tilt, Nix, GenAI

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## Experience

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### ENCORA Inc. Technical Lead Engineer 2025-May 2026

1. Project **Avid Content Core** - Avid Content Core is the cloud-based production solution, combining modern, cloud-native collaboration with Avid's trusted production backbone for transcoding, archiving, and orchestration.
  - Ownership of Python module which is based on clean architecture and EKS based microservices.
  - Contributed to initial setup, architecture, communication with other services and manifestation.
  - Worked on Postgres database setup and manifest repo-based service development.

### CIENA Engineer 2B 2022-2025

1. Project **Blueplanet** - A centralized user access control application based on Python-Django microservice which enables multitude of authentication systems like OAuth, JWT, SAML, LDAP with RESTful APIs and alert system with AWS SES.
  - Led ownership of module which comprises coding, versioning, unit-test, installation, debugging.
  - Developed RESTful APIs using Django Rest Framework and wrote test cases using PyTest.
  - Integrating, packaging and deploying on K8 environments using Helm, docker and Kubernetes.

### AUJAS CYBERSECURITY Senior Security Engineer 2018-2022

1. Project **Appliances** - A Raspberry Pi based Python application which provides packet analysis, intrusion detection and prevention, updating malware database, assess vulnerabilities and provides RESTful APIs for data, and providing alerts. The application is based on Ubuntu on Raspberry Pi to act as a node in a private network.
  - Contributed to design, POC, coding, security modelling, software update and authentication.
  - Modelled secure RESTful communication among Raspberry Pi, mobile apps and EC2 backend server.
  - Designed Scapy and Snort based packet analysis system integrated with malware detection and blocking.
  - Designed to fetch/update malware database from Google safe-browsing and other sources.
  - Developed module to control network filtering using IPtables, IPset, NetFilterQueue.
  - Developed web module to provide RESTful connection between EC2 server and Raspberry Pi.
2. Project **WhiteHaX** - A cloud-hosted, automated, cyber-readiness verification (pen-testing) platform.
  - Developed this web application which provides verification of cyber-readiness by simulating several threat scenarios, deployed security infrastructure, including network perimeter defences and endpoint security & controls.
  - Developed security vulnerability scanner system to generate security health of the user system.
  - Developed central backend system to initiate scanning, processing data and generating report.
  - Added new potential attacks(threats) scenarios for the application.
  - Developed self-destructing executable version of this web-based scanner using Python.
  - Designed database for security scan profiling and scan data dumping.
  - Did setup of Production and Development environments use AWS EC2.
  - Created multitude of security assessment profile with Python library assesses like malware response of the system, DLP policy of existing security system, password vulnerability, privilege user vulnerability, etc.

### IQVIA Senior Software Engineer 2015-2018

1. Project **Centralized Monitoring**: An alerts-based system which collects data from drugs studies and provide a dashboard to medical practitioner to access alerts and analytic of data and control section to perform actions.
  - Developed and enhanced new and existing features in the webapp based on Django and Postgres.
  - Performed databased design changes, wrote SQL queries and written Django models.
  - Performed code debug, log debug on different env, supported code deployment on EC2.
  - Performed updating new features in design doc, code review, code repo management.

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## Education

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Bachelor of Technology, Computer Science & Engineering

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## Certifications

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- Certified in Cybersecurity (cc) by ISC2
  - Issued by International Information System Security Certification Consortium.
- Python certification by FOSSEE
  - MHRD, IIT Bombay
- Paper published:
  - Mines Explorer Cloud Networked Robot, IJETAE- Volume 3, Issue 10
  - 10/2013, ©copyrighted with IJETAE (ISSN 2250-2459, ISO 9001:2008 Certified Journal)